

how does .splice method work in javascript

Ah, `.splice()` —powerful, a little dangerous if you're not watching closely 😊

Here's the clear, practical rundown.

What `.splice()` does

`Array.prototype.splice()` **changes an array in place** by:

- removing elements
- inserting new elements
- or both

Basic syntax

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```
array.splice(start, deleteCount, item1, item2, ...)
```

Parameters

1. `start` – index where changes begin
2. `deleteCount` – how many items to remove
3. `item1, item2, ...` (*optional*) – items to insert at `start`

Removing elements

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```
let arr = ["a", "b", "c", "d"];
```

```
arr.splice(1, 2);
```

```
console.log(arr);
```

Result:

js

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```
["a", "d"]
```

Explanation:

- Start at index 1
 - Remove 2 items ("b" and "c")
-

Inserting elements (without deleting)

Set deleteCount to 0:

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```
let arr = [1, 2, 4, 5];
```

```
arr.splice(2, 0, 3);
```

```
console.log(arr);
```

Result:

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```
[1, 2, 3, 4, 5]
```

Replacing elements

Remove some, insert some:

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```
let arr = ["red", "green", "blue"];
```

```
arr.splice(1, 1, "yellow");
```

```
console.log(arr);
```

Result:

js

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```
["red", "yellow", "blue"]
```

Return value (important!)

`.splice()` returns the removed elements:

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```
let arr = [10, 20, 30, 40];

let removed = arr.splice(1, 2);

console.log(removed); // [20, 30]
console.log(arr);     // [10, 40]
```

Negative indexes

Negative `start` counts from the end:

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```
let arr = ["a", "b", "c", "d"];

arr.splice(-2, 1);
console.log(arr);
```

Result:

js

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```
["a", "b", "d"]
```

Key things to remember ⚠️

- ✅ Mutates the original array
 - ❌ Not safe if you need immutability (e.g., React state)
 - 🆚 Don't confuse with `.slice()` :
 - `.slice()` → non-mutating
 - `.splice()` → mutating
-

If you want, I can:

- compare `.splice()` vs `.slice()` side-by-side
- show real-world examples (todo lists, carts, etc.)
- explain how to avoid mutation using modern JS

Just say the word 🗨️

